

COMSATS University Islamabad (Lahore Campus)

Department of Computer Engineering

Centralized Sprint Plan

Industrial Safety Intelligence Platform

5 Months · 5 Sprints · 20 Weeks

Master Timeline — Both Members

Duration: 5 months (20 weeks)

Sprint length: 4 weeks each

Version: 3.0 (compressed from 8-month plan)

Member A plan: `member_a_sprint_plan.tex`

Member B plan: `member_b_sprint_plan.tex`

Spring–Fall 2026

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1 How to Use This Document

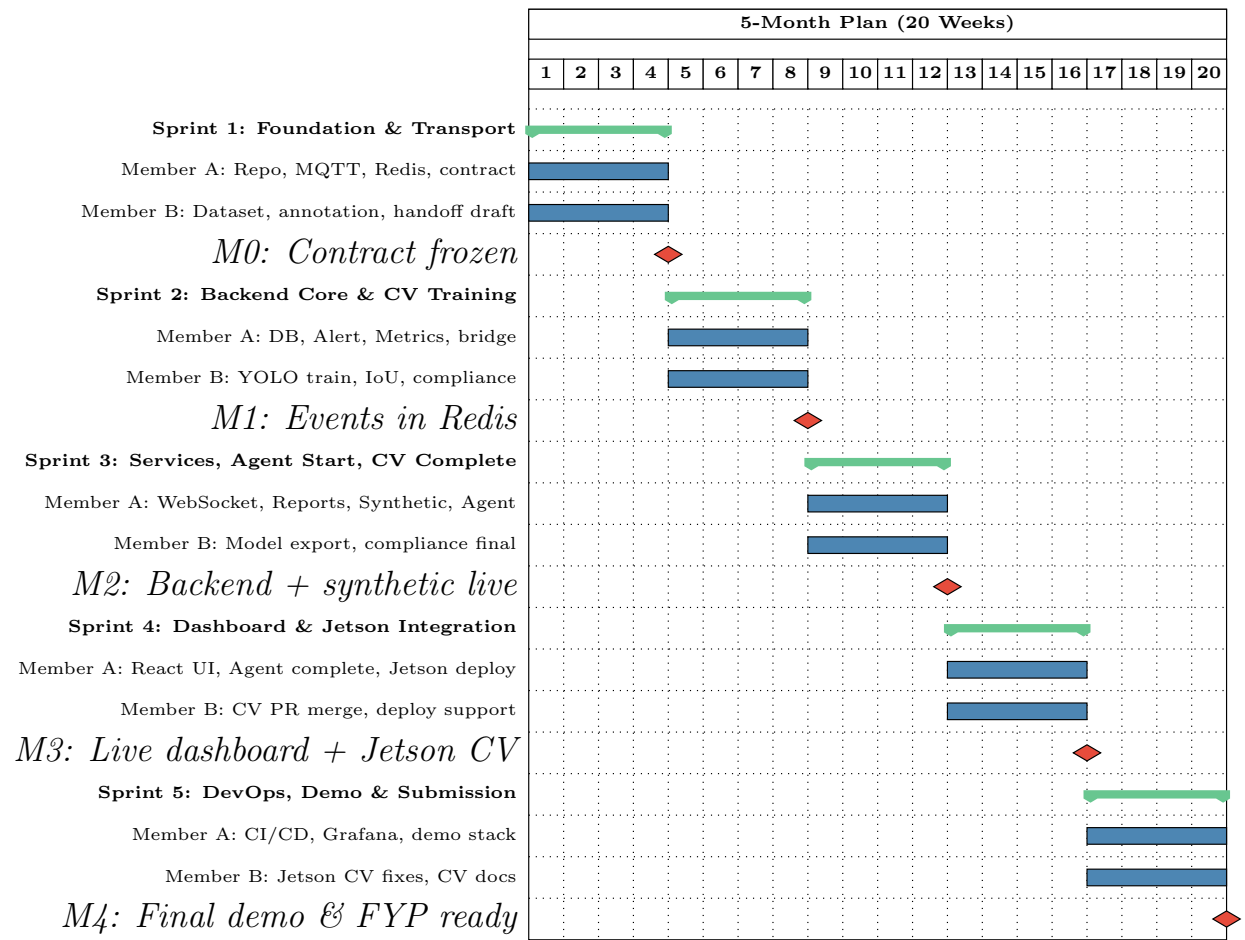
- **This file** is the single source of truth for **sprint dates, milestones, and joint sessions**.
- **Member A** follows `member_a_sprint_plan.tex` for platform, MQTT, deploy, and DevOps tasks only.
- **Member B** follows `member_b_sprint_plan.tex` for CV development tasks only.
- When dates or milestones change, update **this file first**, then sync member files.
- Track tickets in **GitHub Projects** (see `README.md` in this folder).

2 Team Roles (Summary)

Member	Owns
A	Full MQTT pipeline, Jetson deploy, 5 microservices, agent, dashboard, synthetic data, DevOps
B	YOLO inference, compliance engine, evidence capture, CV handoff dict — no MQTT
Joint	Frozen contract, handoff interface, integration sessions, final demo

Detection classes (supervisor-approved): worker, helmet, vest, fire, smoke, spill. See `DETECTION_CLASS_SCOPE.md`.

3 5-Month Timeline (Gantt)



4 Sprint 1 (Weeks 1–4): Foundation & Transport

Sprint 1 — Lock the contract; start both tracks

Sprint goal

Monorepo live, frozen event contract signed, MQTT→Redis path working with simulator; Member B starts dataset and handoff spec.

Member A deliverables

- Repo scaffold, docker-compose.yml skeleton, dev environment
- contracts/events.py drafted; Mosquitto + Redis in Compose
- MQTT-Redis bridge + event simulator publishing valid events

Member B deliverables

- Dev environment, GPU verified, annotation tool ready
- edge/inference/README.md with 6 supervisor classes

- Draft CV handoff interface (Python dict schema) for Member A

Collaboration Touchpoint

Week 4 — Integration session #1

- Both sign off on `contracts/events.py` and handoff interface (M0)
- Member B provides sample handoff JSON; Member A validates against contract
- Sprint review: simulator events visible in Redis

5 Sprint 2 (Weeks 5–8): Backend Core & CV Training

Sprint 2 — Data layer + first microservices + YOLO training

Sprint goal

Alert and Metrics services consuming Redis Streams; PostgreSQL + MinIO live; Member B training YOLO on 6 classes.

Member A deliverables

- PostgreSQL + TimescaleDB schema, MinIO evidence bucket
- Alert Service (:8001), Metrics Service (:8002)
- GET `/api/alerts`, GET `/api/metrics/compliance`

Member B deliverables

- 500+ annotated images (minimum); YOLOv8n training in progress
- IoU association + compliance engine prototype
- Evidence frame save returning `image_path`

Collaboration Touchpoint

Week 8 — Integration session #2 (M1)

- Simulator (or stub handoff) events persist in `violations` table
- Member B demo: inference + handoff dict on laptop (no MQTT)
- Carry-over list for Sprint 3 documented in GitHub Issues

6 Sprint 3 (Weeks 9–12): Platform Services & CV Completion

Sprint 3 — Real-time layer

Sprint goal

WebSocket live; 90-day synthetic seed; reporting PDF; agent graph started; Member B model export ready.

Member A deliverables

- WebSocket Gateway (:8004), Reporting Service (:8003)
- Synthetic data engine — `90day_baseline` seed
- Agent Service (:8005) — LangGraph skeleton + 3 tools

Member B deliverables

- Trained model $\text{mAP}@0.5 \geq 0.70$ on test set
- ONNX export + compliance engine complete
- Merged PRs to `edge/inference/`, `edge/compliance_engine/`

Collaboration Touchpoint

Week 12 — Integration session #3 (M2)

- Dashboard can query synthetic history via REST
- Member B handoff dict consumed by Member A's test harness (still no B-side MQTT)
- Agent responds to one test query

7 Sprint 4 (Weeks 13–16): Dashboard & Jetson Integration

Sprint 4 — Full UI + deploy CV on Jetson + agent modes

Sprint goal

React dashboard with WebSocket; agent reactive + chat modes; Member A deploys B's CV + MQTT publisher on Jetson.

Member A deliverables

- React dashboard: gauge, alerts, analytics, agent chat, zone config, reports

- MQTT publisher wrapping B's handoff; Jetson flash + deploy
- Agent: RAG + proactive mode; full REST surface

Member B deliverables

- Final CV fixes from Member A's deploy feedback
- edge/README.md + model weights delivery package
- Support integration sessions (explain code, tune thresholds)

Collaboration Touchpoint

Week 16 — Integration session #4 (M3)

- End-to-end: Jetson CV → MQTT → dashboard gauge updates
- Helmet-off demo triggers alert + optional agent briefing
- Both attend; blockers resolved before Sprint 5

8 Sprint 5 (Weeks 17–20): DevOps, Demo & Submission

Sprint 5 — Production polish

Sprint goal

One-command `docker compose up`; CI/CD green; Grafana dashboards; 12-minute demo rehearsed; documentation complete.

Member A deliverables

- Full Compose stack, GitHub Actions CI/CD
- Prometheus + Grafana (System Health + Safety KPIs)
- Disable simulator; live Jetson pipeline for demo

Member B deliverables

- CV section of report + viva prep (model, IoU, compliance, handoff)
- Final model benchmark on Jetson (FPS, accuracy note)
- Review joint presentation slides

Collaboration Touchpoint**Week 20 — Final demo & submission (M4)**

- 12-minute demo script rehearsed twice
- All GitHub Issues for Sprint 5 closed or deferred with supervisor note
- FYP report, slides, and code tag `v1.0-demo`

9 Milestones

ID	Week	Criteria (both must verify)
M0	4	Contract + handoff interface signed
M1	8	MQTT pipeline → Redis → DB writes
M2	12	Synthetic data + WebSocket + B's model PR merged
M3	16	Jetson live → dashboard real-time
M4	20	Full demo + submission ready

10 Standing Collaboration Calendar

Event	When	Who
Sprint planning	Week 1 of each sprint	Both
Weekly sync	Once per week	Both
Sprint review	Week 4 of each sprint	Both
Integration session	Weeks 4, 8, 12, 16, 20	Both
Sprint log due	Every Sunday 11:59 PM	Each member

11 Risk Mitigation (5-Month Compression)

- **Parallel work:** B trains CV while A builds platform — do not wait for Jetson until Sprint 4.
- **Simulator:** Member A's event simulator unblocks all backend sprints without B hardware.
- **Scope freeze:** 6 detection classes only; no goggles/forklift.
- **Slip buffer:** Optional gloves/boots dropped if Sprint 3 training behind schedule.
- **Integration weeks:** Non-negotiable — missing two in a row triggers supervisor escalation.